longitude, depth of penetration). Together with time, this is therefore an example of a universe with *seven* spacetime dimensions.

Now comes a leap. Although it is impossible to draw, imagine that at every point in the three extended dimensions of everyday life, the universe has not one extra dimension as in Figure 12.7, not two extra dimensions as in Figure 12.8a, not three extra dimensions as in Figure 12.8b, but six extra space dimensions. I certainly can't visualize this and I've never met anyone who can. But its meaning is clear. To specify the spatial location of a Planck-sized worm in such a universe requires *nine* pieces of information: three to locate its position in the usual extended dimensions and six more to locate its position in the curled-up dimensions tacked on to that point. When time is also taken into account, this is a ten-spacetime-dimensional universe, as required by the equations of string theory. If the extra six dimensions are curled up small enough, they would easily have escaped detection.

The Shape of Hidden Dimensions

The equations of string theory actually determine more than just the number of spatial dimensions. They also determine the kinds of shapes the extra dimensions can assume.[18] In the figures above, we focused on the simplest of shapes—circles, hollow spheres, solid balls—but the equations of string theory pick out a significantly more complicated class of six-dimensional shapes known as Calabi-Yau shapes or Calabi-Yau spaces. These shapes are named after two mathematicians, Eugenio Calabi and Shing-Tung Yau, who discovered them mathematically long before their relevance to string theory was realized; a rough illustration of one example is given in Figure 12.9a. Bear in mind that in this figure a two-dimensional graphic illustrates a six-dimensional object, and this results in a variety of significant distortions. Even so, the picture gives a rough sense of what these shapes look like. If the particular Calabi-Yau shape in Figure 12.9a constituted the extra six dimensions in string theory, on ultramicroscopic scales space would have the form illustrated in Figure 12.9b. As the Calabi-Yau shape would be tacked on to every point in the usual three dimensions, you and I and everyone else would right now be surrounded by and filled with these little shapes. Literally, as you walk from one place to another, your body would move through